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A Novel Proof of the Desnanot-Jacobi Determinant Identity

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In addition to writing the book *Alice in Wonderland*, Charles Lutwidge Dodgson devised an algorithm for computing determinants of square integer matrices called the *method of contractants*. This method is especially useful when computing determinants of integer matrices by hand because all intermediate calculations use only integers. However, the proof that this method works relies heavily on the *Desnanot-Jacobi Determinant identity* [2]. We prove this identity using induction, basic matrix algebra, and differential calculus (techniques known to most undergraduate students of mathematics).

Preliminaries

Let $\mathbb{A}$ be an $n \times n$ matrix with entries $a_{i,j}$ for $1 \le i, j \le n$ and *determinant* $|\mathbb{A}|$. We denote by $\mathbb{A}_{i|k}$ the matrix $\mathbb{A}$ with row i and column k deleted; similarly let $\mathbb{A}_{i,j|k,\ell}$ be the matrix obtained by deleting rows i and j *and* columns k and ℓ:

$$\mathbb{A}_{i|k} := \begin{bmatrix} & a_{k,1} & \\ a_{i,1} & & a_{i,n} \\ & a_{k,n} & \end{bmatrix}, \quad \mathbb{A}_{i,j|k,\ell} := \begin{bmatrix} & a_{k,1} & a_{\ell,1} & \\ a_{i,1} & & & a_{i,n} \\ a_{j,1} & & & a_{j,n} \\ & a_{k,n} & a_{\ell,n} & \end{bmatrix}.$$

With this notation, for example, the determinant of $\mathbb{A}$ using *cofactor expansion* along the ith row is given by:

$$|\mathbb{A}| = \sum_{j=1}^{n} (-1)^{i+j} a_{i,j} \cdot |A_{i|j}|$$

where $(-1)^{i+j}|A_{i,j}|$ is the so-called *cofactor* of $a_{i,j}$.

We prove the following proposition:

Proposition 1 (The Desnanot-Jacobi Identity)**.** For a square matrix $\mathbb{A}$,

$$|\mathbb{A}| \cdot \left|\mathbb{A}_{i,j|k,\ell}\right| = \left|\mathbb{A}_{i|k}\right| \cdot \left|\mathbb{A}_{j|\ell}\right| - \left|\mathbb{A}_{i|\ell}\right| \cdot \left|\mathbb{A}_{j|k}\right|. \tag{1}$$

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Example 1. Let $(i, j, k, \ell) = (1, 3, 2, 4)$ and

$$\mathbb{A} = \begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 0 & -4 & 3 \\ 0 & -3 & -3 & -1 \\ 3 & -2 & 4 & -3 \end{bmatrix}.$$

We have

$$\mathbb{A}_{1,3|2,4} = \begin{bmatrix} 1 & -4 \\ 3 & 4 \end{bmatrix}, \quad \mathbb{A}_{1|2} = \begin{bmatrix} 1 & -4 & 3 \\ 0 & -3 & -1 \\ 3 & 4 & -3 \end{bmatrix}, \quad \mathbb{A}_{1|4} = \begin{bmatrix} 1 & 0 & -4 \\ 0 & -3 & -3 \\ 3 & -2 & 4 \end{bmatrix},$$

$$\mathbb{A}_{3|4} = \begin{bmatrix} 2 & -1 & -3 \\ 1 & 0 & -4 \\ 3 & -2 & 4 \end{bmatrix}, \qquad \mathbb{A}_{3|2} = \begin{bmatrix} 2 & -3 & -1 \\ 1 & -4 & 3 \\ 3 & 4 & -3 \end{bmatrix}.$$

Notice that $|\mathbb{A}| \cdot \left|\mathbb{A}_{1,3|2,4}\right| = (-156)(16) = -2496$ and

$$\left|\mathbb{A}_{1|2}\right| \cdot \left|\mathbb{A}_{1|4}\right| - \left|\mathbb{A}_{3|4}\right| \cdot \left|\mathbb{A}_{3|2}\right| = (52)(-54) - (6)(-52) = -2496,$$

as Proposition 1 implies.

Proof by induction

To prove equation (1), we proceed by induction. Since only the sign of the determinant changes when two rows (or columns) are interchanged, it suffices to show that

$$|\mathbb{A}| \cdot \left|\mathbb{A}_{1,2|1,2}\right| = \left|\mathbb{A}_{1|1}\right| \cdot \left|\mathbb{A}_{2|2}\right| - \left|\mathbb{A}_{1|2}\right| \cdot \left|\mathbb{A}_{2|1}\right| \tag{2}$$

as we can just swap row i with row 1, row k with row 2, column j with column 1, and column k with column 2. Four interchanges (an even number) has no effect on the determinant.

Base case It is easy to verify equation (2) explicitly when $n = 2$, provided we utilize the fact that the determinant of the 0×0 matrix is one.

$$\begin{vmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{vmatrix} \cdot 1 = a_{2,2} \cdot a_{1,1} - a_{2,1} \cdot a_{1,2} = \left|\mathbb{A}_{1|1}\right| \cdot \left|\mathbb{A}_{2|2}\right| - \left|\mathbb{A}_{1|2}\right| \cdot \left|\mathbb{A}_{2|1}\right|.$$

Remark 1. Some readers may find it perplexing that the determinant of the 0×0 matrix "[ ]" is one. Intuitively, if we were to do cofactor expansion along the 1×1 matrix $[a_{1,1}]$, then we compute $|a_{1,1}| = a_{1,1} \cdot |\mathbb{A}_{1|1}| = a_{1,1} \cdot |[\ \ ]|$. Given that $|a_{1,1}| = a_{1,1}$ it follows $|[\ \ ]| = 1$.

This is also apparent when using the Leibniz formula for determinants. [4] Here

$$|\mathbb{A}| = \sum_{\tau \in S_n} \operatorname{sgn}(\tau) \prod_{i=1}^{n} a_{i,\tau(i)},$$

where S_n is the group of all permutations of n indices and sgn is the signature of permutation $\tau \in S_n$. There is only one permutation of an empty set (the identity permutation, e) and the empty product is equal to one. Thus

$$|\mathbb{A}| = \sum_{\tau \in S_0} \operatorname{sgn}(\tau) \prod_{i=1}^{0} a_{i,\tau(i)} = \sum_{\tau \in \{e\}} 1 \cdot 1 = 1,$$

when $\mathbb{A}$ is 0×0.

Inductive step Assume that equation (2) is correct for an $n \times n$ matrix $\mathbb{A}$. Next, extend $\mathbb{A}$ by one row and one column and denote this new $(n+1) \times (n+1)$ matrix by $\mathbb{A}^+$. We need to prove that

$$\left|\mathbb{A}^+\right| \cdot \left|\mathbb{A}^+_{1,2|1,2}\right| = \left|\mathbb{A}^+_{1|1}\right| \cdot \left|\mathbb{A}^+_{2|2}\right| - \left|\mathbb{A}^+_{1|2}\right| \cdot \left|\mathbb{A}^+_{2|1}\right| \tag{3}$$

to conclude the main result.

Note that $|\mathbb{A}|$ is a polynomial in n^2 variables with $n!$ terms, each comprised of a product of n *distinct* $a_{i,j}$'s.

For instance, when $\mathbb{A}$ is a 3×3 matrix, we have

$$\begin{aligned} |\mathbb{A}| &= a_{1,1}a_{2,2}a_{3,3} + a_{1,2}a_{2,3}a_{3,1} + a_{1,3}a_{2,1}a_{3,2} \\ &\quad - a_{1,1}a_{2,3}a_{3,2} - a_{1,2}a_{2,1}a_{3,3} - a_{1,3}a_{2,2}a_{3,1}. \end{aligned}$$

This polynomial has nine variables and six terms. Notice that there are no powers greater than one, and therefore the expression is linear in each $a_{i,j}$.

Regarding $|\mathbb{A}|$ as a polynomial means that the coefficient (or cofactor) to $a_{i,j}$ is equal to $\frac{\partial}{\partial a_{i,j}} |\mathbb{A}|$. Accordingly, the usual cofactor expansion along the jth column of $\mathbb{A}$, namely

$$|\mathbb{A}| = \sum_{i=1}^{n} a_{i,j} \cdot (-1)^{i+j} \left|\mathbb{A}_{i|j}\right|, \tag{4}$$

is also given by

$$|\mathbb{A}| = \sum_{i=1}^{n} a_{i,j} \cdot \frac{\partial\, |\mathbb{A}|}{\partial a_{i,j}}. \tag{5}$$

We now define, for each $1 \le i \le n$, a linear differential operator $D^{(i)}$ with constant coefficients by

$$D^{(i)} := \sum_{j=1}^{n} a_{n+1,j} \cdot \frac{\partial}{\partial a_{i,j}}, \tag{6}$$

where $a_{n+1,j}$ (for $j = 1 \ldots n$) are the first n elements of the last row of $\mathbb{A}^+$. Applied to $|\mathbb{A}|$, or any of the other five determinants of equation (2), this operator replaces the ith row of $\mathbb{A}$ (or $\mathbb{A}_{i,j|k,\ell}, \ldots$) by the last row of $\mathbb{A}^+$ (or $\mathbb{A}^+_{i,j|k,\ell}, \ldots$), short of the $a_{n+1,n+1}$ "corner" element, and returns the correspondingly modified determinant. It follows that

$$-D^{(i)}|\mathbb{A}| = |\mathrm{swaprow}(\mathbb{A}^+;\ i,\ n+1)_{n+1|n+1}|,$$

and therefore this is the cofactor of $a_{i,n+1}$ in $\mathbb{A}^+$.

Example 2. Let $\mathbb{A}$ be a 2×2 matrix, so that

$$\mathbb{A} = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \quad \text{and} \quad \mathbb{A}^+ = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}.$$

For $D^{(2)} = a_{3,1}\frac{\partial}{\partial a_{2,1}} + a_{3,2}\frac{\partial}{\partial a_{2,2}}$, we have

$$D^{(2)}|\mathbb{A}| = D^{(2)}(a_{1,1}a_{2,2} - a_{1,2}a_{2,1}) = -a_{3,1}a_{1,2} + a_{3,2}a_{1,1} = \begin{vmatrix} a_{1,1} & a_{1,2} \\ a_{3,1} & a_{3,2}, \end{vmatrix}$$

which is the determinant of $\mathbb{A}$ with its second row replaced by the last row of $\mathbb{A}^+$. Moreover, $-D^{(2)}|\mathbb{A}|$ is indeed the cofactor of $a_{2,3}$.

Cofactor expansion of $\mathbb{A}^+$ along the last column gives

$$\left|\mathbb{A}^+\right| = a_{n+1,n+1}\left|\mathbb{A}\right| - \sum_{i=1}^{n} a_{i,n+1} D^{(i)}\left|\mathbb{A}\right|, \tag{7}$$

and similarly for any other matrix of equation (3). We can thus expand all the determinants of equation (3) using equation (7) and obtain

$$\begin{aligned} &\left(a_{n+1,n+1}\left|\mathbb{A}\right| - \sum_{i=1}^{n} a_{i,n+1} D^{(i)}\left|\mathbb{A}\right|\right) \\ &\qquad \times \left(a_{n+1,n+1}\left|\mathbb{A}_{1,2|1,2}\right| - \sum_{i=1}^{n} a_{i,n+1} D^{(i)}\left|\mathbb{A}_{1,2|1,2}\right|\right) \\ &= \left(a_{n+1,n+1}\left|\mathbb{A}_{1|1}\right| - \sum_{i=1}^{n} a_{i,n+1} D^{(i)}\left|\mathbb{A}_{1|1}\right|\right) \\ &\qquad \times \left(a_{n+1,n+1}\left|\mathbb{A}_{2|2}\right| - \sum_{i=1}^{n} a_{i,n+1} D^{(i)}\left|\mathbb{A}_{2|2}\right|\right) \\ &\qquad - \left(a_{n+1,n+1}\left|\mathbb{A}_{1|2}\right| - \sum_{i=1}^{n} a_{i,n+1} D^{(i)}\left|\mathbb{A}_{1|2}\right|\right) \\ &\qquad \times \left(a_{n+1,n+1}\left|\mathbb{A}_{2|1}\right| - \sum_{i=1}^{n} a_{i,n+1} D^{(i)}\left|\mathbb{A}_{2|1}\right|\right). \end{aligned} \tag{8}$$

In order to simplify equation (8), notice first that

$$D^{(k)}D^{(i)}\left|\mathbb{A}\right| = 0 \tag{9}$$

not only for $\mathbb{A}$, but also for the other five determinants of the induction hypothesis. This is because, when $k \neq i$, the left-hand side of equation (9) results in a determinant of a matrix with two identical rows; when $k = i$ the second $D^{(i)}$ is differentiating with respect to $a_{i,j}$ which is no longer a part of $D^{(i)}\left|\mathbb{A}\right|$. This implies (by the product rule for differentiation) that

$$D^{(k)}D^{(i)}\left|\mathbb{A}\right|\left|\mathbb{A}_{1,2|1,2}\right| = D^{(k)}\left|\mathbb{A}\right| \cdot D^{(i)}\left|\mathbb{A}_{1,2|1,2}\right| + D^{(i)}\left|\mathbb{A}\right| \cdot D^{(k)}\left|\mathbb{A}_{1,2|1,2}\right| \tag{10}$$

for a product of any two of the six determinants in equation (2).

We demonstrate equation (8) by expanding both sides and collecting terms proportional to $a^2_{n+1,n+1}$, $a_{n+1,n+1}a_{i,n+1}$, and finally to $a_{i,n+1}a_{j,n+1}$. The terms proportional to $a^2_{n+1,n+1}$ are $|\mathbb{A}| \cdot \left|\mathbb{A}_{1,2|1,2}\right| = \left|\mathbb{A}_{1|1}\right| \cdot \left|\mathbb{A}_{2|2}\right| - \left|\mathbb{A}_{1|2}\right| \cdot \left|\mathbb{A}_{2|1}\right|$, which is trivially true by the induction hypothesis.

The terms proportional to $a_{n+1,n+1}a_{i,n+1}$ are

$$\begin{aligned}
&- D^{(i)} |\mathbb{A}| \cdot \left|\mathbb{A}_{1,2|1,2}\right| - |\mathbb{A}| \cdot D^{(i)} \left|\mathbb{A}_{1,2|1,2}\right| \\
&\qquad = D^{(i)} \left|\mathbb{A}_{1|2}\right| \cdot \left|\mathbb{A}_{2|1}\right| + \left|\mathbb{A}_{1|2}\right| \cdot D^{(i)} \left|\mathbb{A}_{2|1}\right| \\
&\qquad\quad - D^{(i)} \left|\mathbb{A}_{1|1}\right| \cdot \left|\mathbb{A}_{2|2}\right| - \left|\mathbb{A}_{1|1}\right| \cdot D^{(i)} \left|\mathbb{A}_{2|2}\right|,
\end{aligned}$$

which is the same as $-D^{(i)}$ applied to the induction hypothesis and thus true. Finally, terms proportional to $a_{i,n+1}a_{j,n+1}$ are

$$\begin{aligned}
&D^{(i)} |\mathbb{A}| \cdot D^{(j)} \left|\mathbb{A}_{1,2|1,2}\right| + D^{(j)} |\mathbb{A}| \cdot D^{(i)} \left|\mathbb{A}_{1,2|1,2}\right| \\
&\qquad = D^{(i)} \left|\mathbb{A}_{1|1}\right| \cdot D^{(j)} \left|\mathbb{A}_{2|2}\right| + D^{(j)} \left|\mathbb{A}_{1|1}\right| \cdot D^{(i)} \left|\mathbb{A}_{2|2}\right| \\
&\qquad\quad - D^{(i)} \left|\mathbb{A}_{1|2}\right| \cdot D^{(j)} \left|\mathbb{A}_{2|1}\right| - D^{(j)} \left|\mathbb{A}_{1|2}\right| \cdot D^{(i)} \left|\mathbb{A}_{2|1}\right|,
\end{aligned}$$

which is the same as $D^{(i)}D^{(j)}$ applied to induction hypothesis and thus true.

Therefore, we have shown that equation (2) implies equation (3) and therefore

$$|\mathbb{A}| \cdot \left|\mathbb{A}_{1,2|1,2}\right| = \left|\mathbb{A}_{1|1}\right| \cdot \left|\mathbb{A}_{2|2}\right| - \left|\mathbb{A}_{1|2}\right| \cdot \left|\mathbb{A}_{2|1}\right| \tag{11}$$

for any square matrix $\mathbb{A}$. It follows that Desnanot-Jacobi Determinant Identity is true.

Finally, we should mention that simpler proofs of this identity have already been published: two by Jacobi himself [5, 6], and the latest (and arguably most elegant) by Bressoud [3]. Bressoud also describes the genesis of this result, discovered for $n = 3$ by Lagrange in 1773 and proven by Desnanot up to $n = 6$ in 1819.

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Summary. We present an alternate proof for the Desnanot-Jacobi identity using induction, basic matrix algebra and elementary differential calculus. It is an instructive demonstration of how it is occasionally necessary to utilize results from different fields of mathematics to solve a specific problem.

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